

Ethanol Blending in Gasoline – Costa Rica

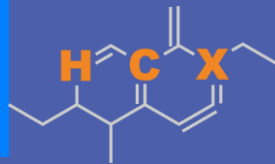
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Ethanol Blending in Latin America

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Latin America to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

Sixteen countries with potential and additional use of ethanol were studied: 1) gasoline market profiles; 2) Optimization of gasoline blends with ethanol and 3) Environmental impact of gasolines blended with ethanol.



Ethanol Blending in Gasoline - Costa Rica



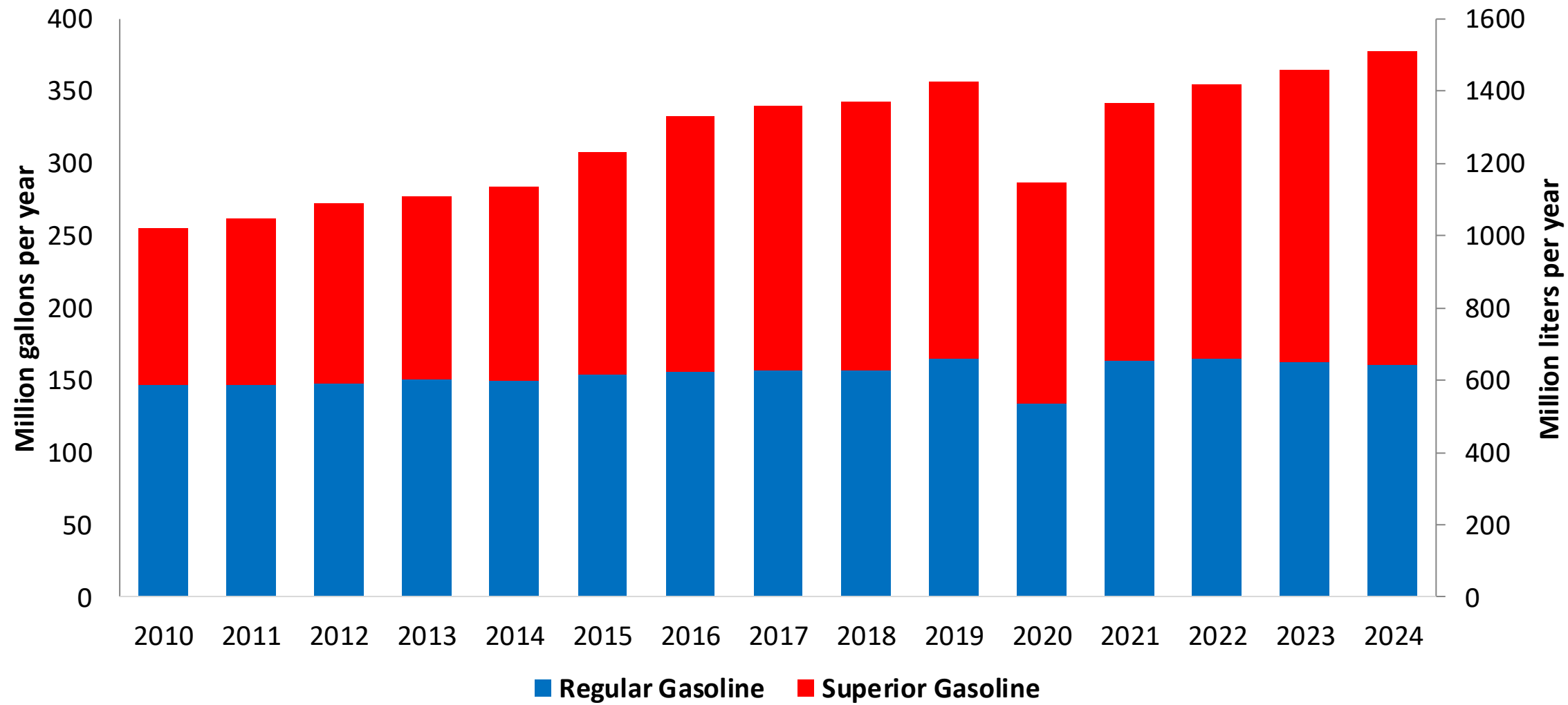
In 2024, gasoline demand was 378 million gallons (1429 million liters). Costa Rica has two gasoline RON 95 and RON 91, with a market share of 45% for RON 91 and 55% for RON 95. The Limon refinery stopped production in 2012. Gasoline supply is done with imports mainly from the United States and some from Europe.

Gasoline blends with up to 10% v/v of ethanol are allowed (INTE E1:2019). However, this mandate is not yet reflected on the gasoline market. Current ethanol consumption is less than 0.2% because there is no stakeholder's consensus for implementation. In 2024, ethanol production was 8.9 million gallons (33 million liters) and ethanol exports mainly to Europe was 7.2 million gallons (27 million liters). Introduction of E10 is expected by the Government in 2026.

Source: Sepse

Gasoline Demand in Costa Rica

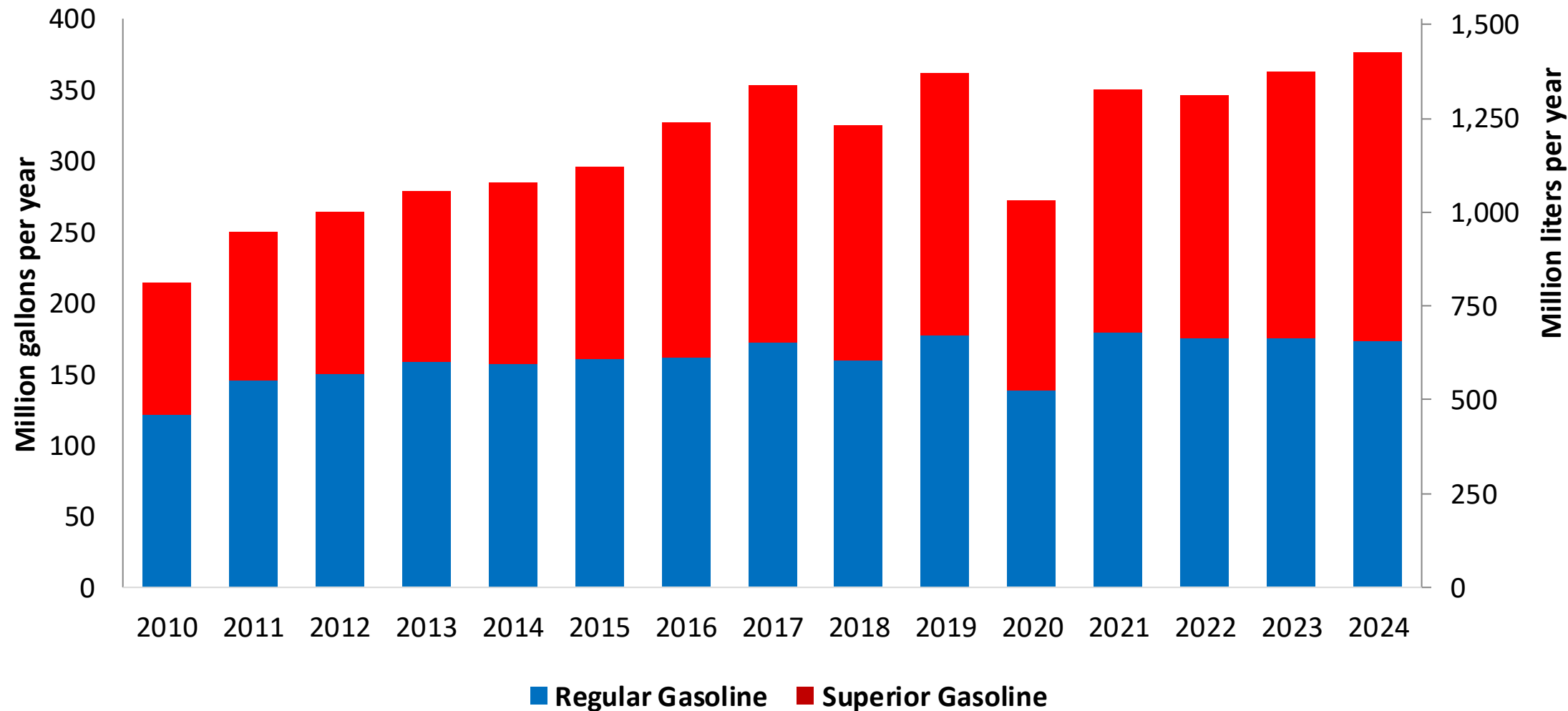
Gasoline Demand by Grade in Costa Rica



Source: Sepse

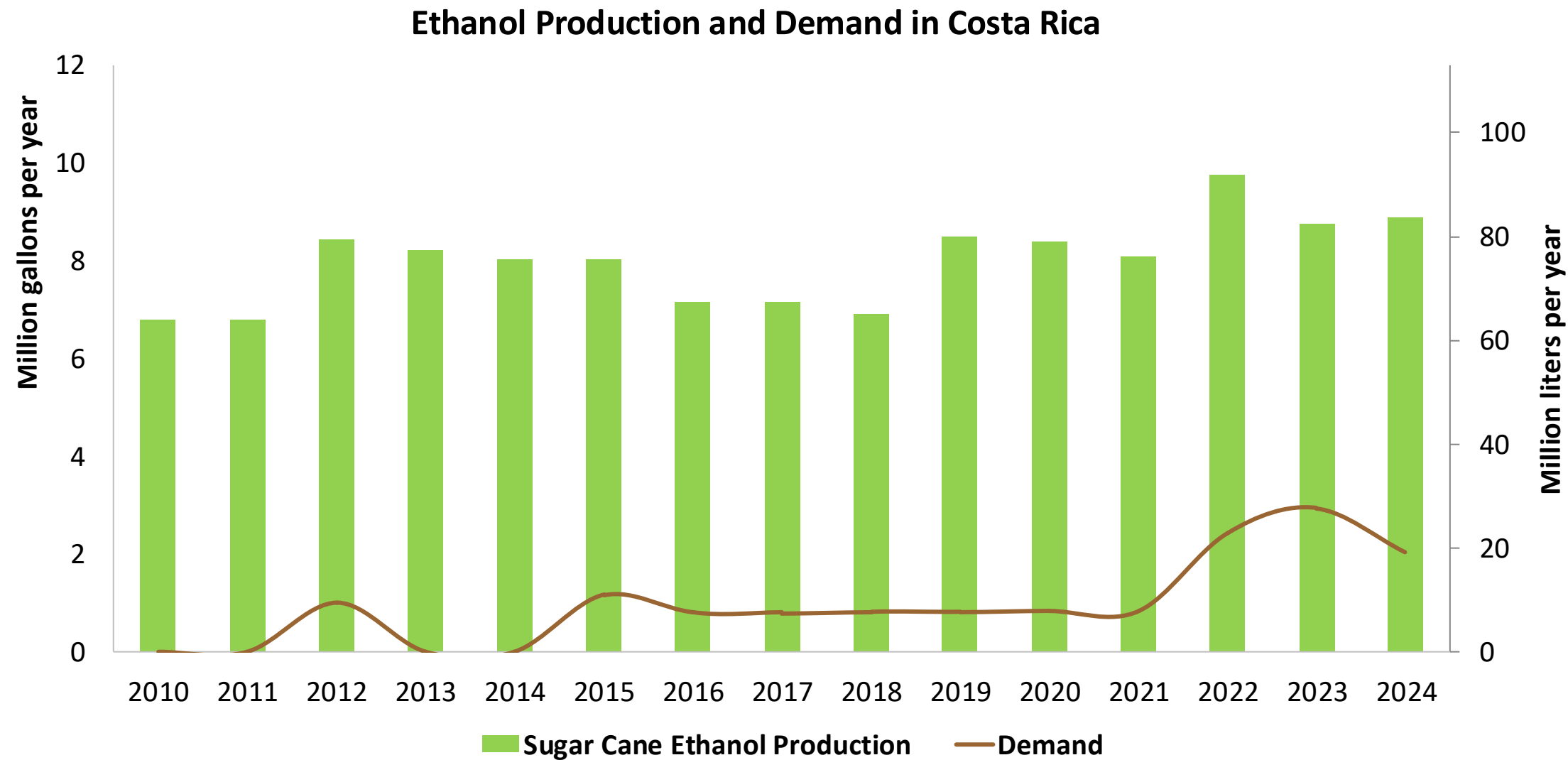
Gasoline Imports in Costa Rica

Gasoline Imports by Grade to Costa Rica



Source: Sepse

Ethanol Demand in Costa Rica



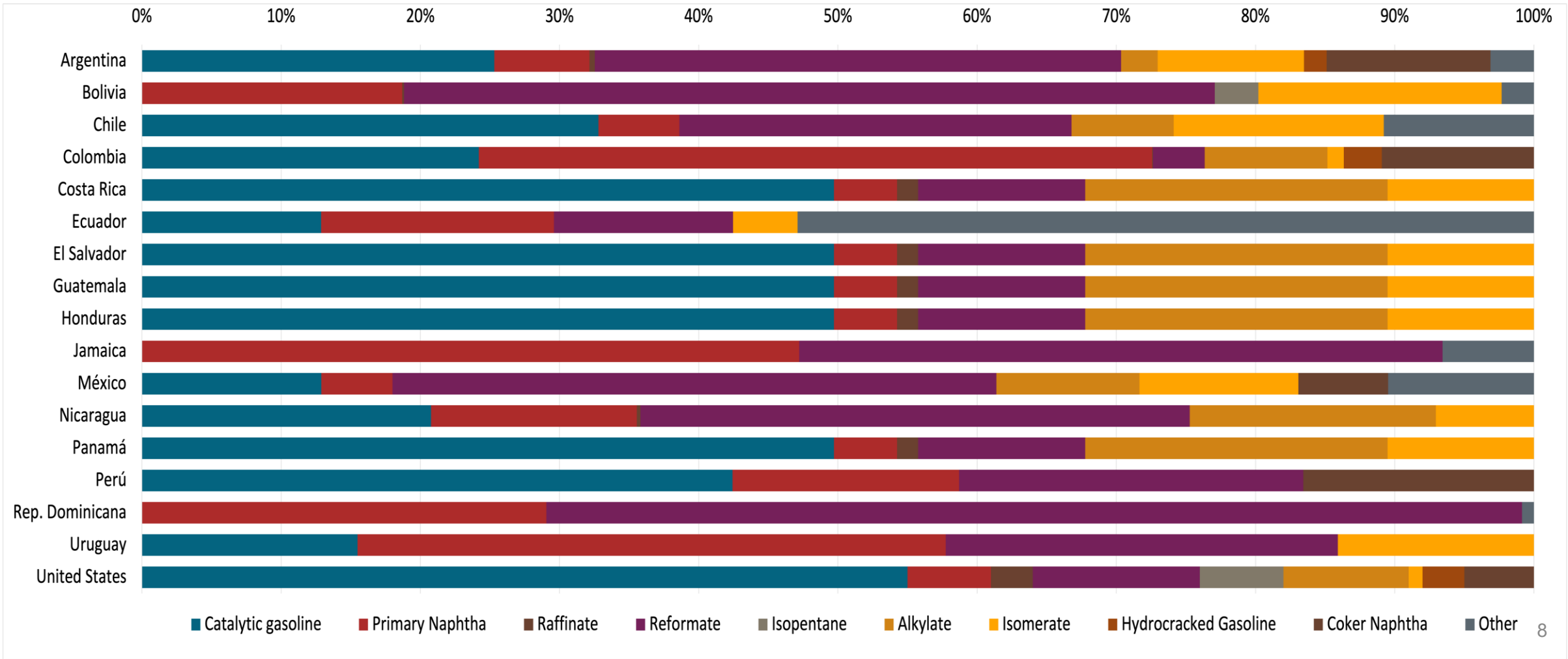
Source: Sepse

Gasoline Quality in Costa Rica

Name	INTE E1:2019		EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2019		2017			
Applicability	Whole country	Whole country	All countries			
Selected Grade	RON 91	RON 95	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 1,5% v/v	< 1,5% v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 35% v/v	< 35% v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	< 18% v/v	< 18% v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 91	> 95	> 95	> 95	> 98	> 98
MON	> 79	> 83	> 85	> 88	> 85	> 88
AKI						
Sulfur Content	< 50 mg/kg	< 50 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	2,7% m/m (3,7% m/m if ethanol is added)	2,7% m/m (3,7% m/m if ethanol is added)	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	< 10% v/v	< 10% v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 69 kPa (< 76 kPa if ethanol is added)	< 69 kPa (< 76 kPa if ethanol is added)	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)						
RVP 37.8°C (Transition)						
MTBE	-		-	-	-	-
Ethers 5 or more C Atoms	-		Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Gasoline Component Blending in Latin America

Gasoline is a blend of a base gasoline and other components. This blending is usually done at blending terminals as only 30% of the world's finished gasoline is distributed directly from refineries. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Latin America are:



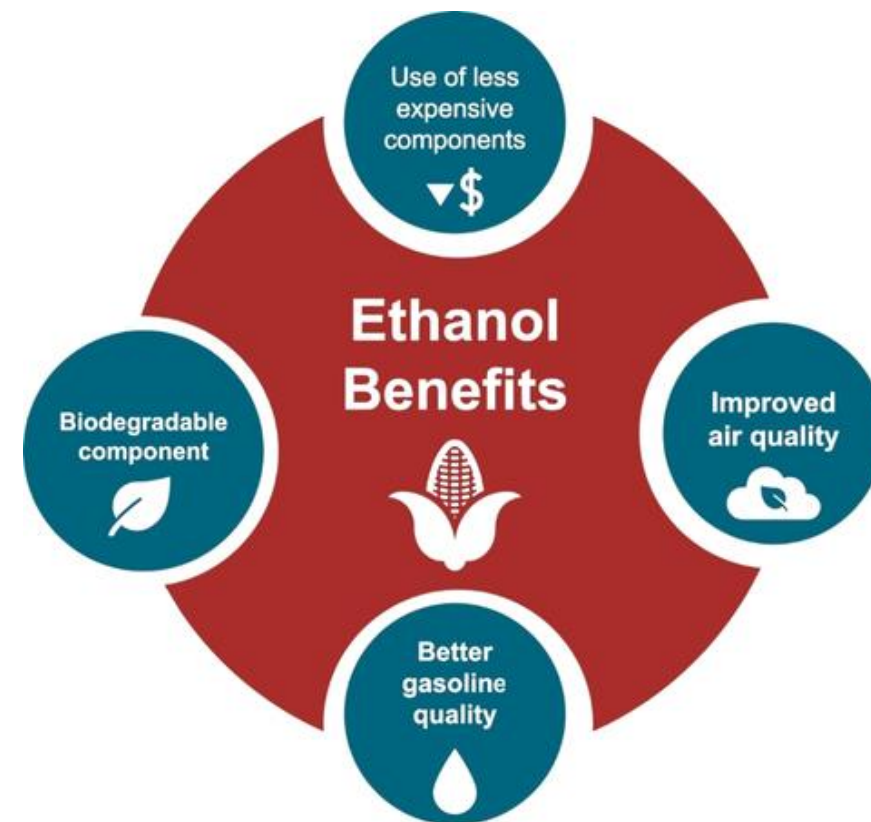
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

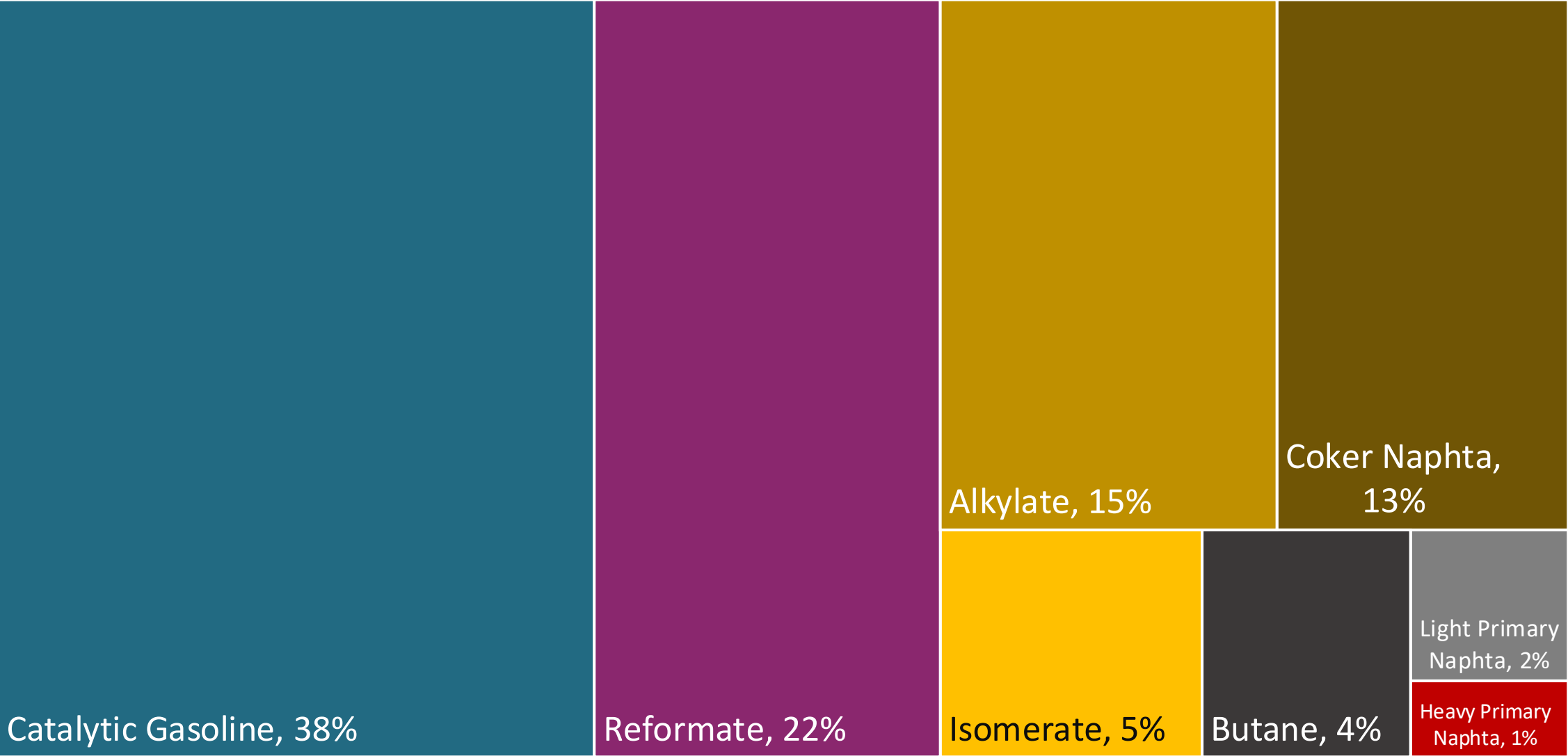
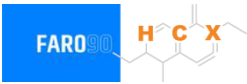
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

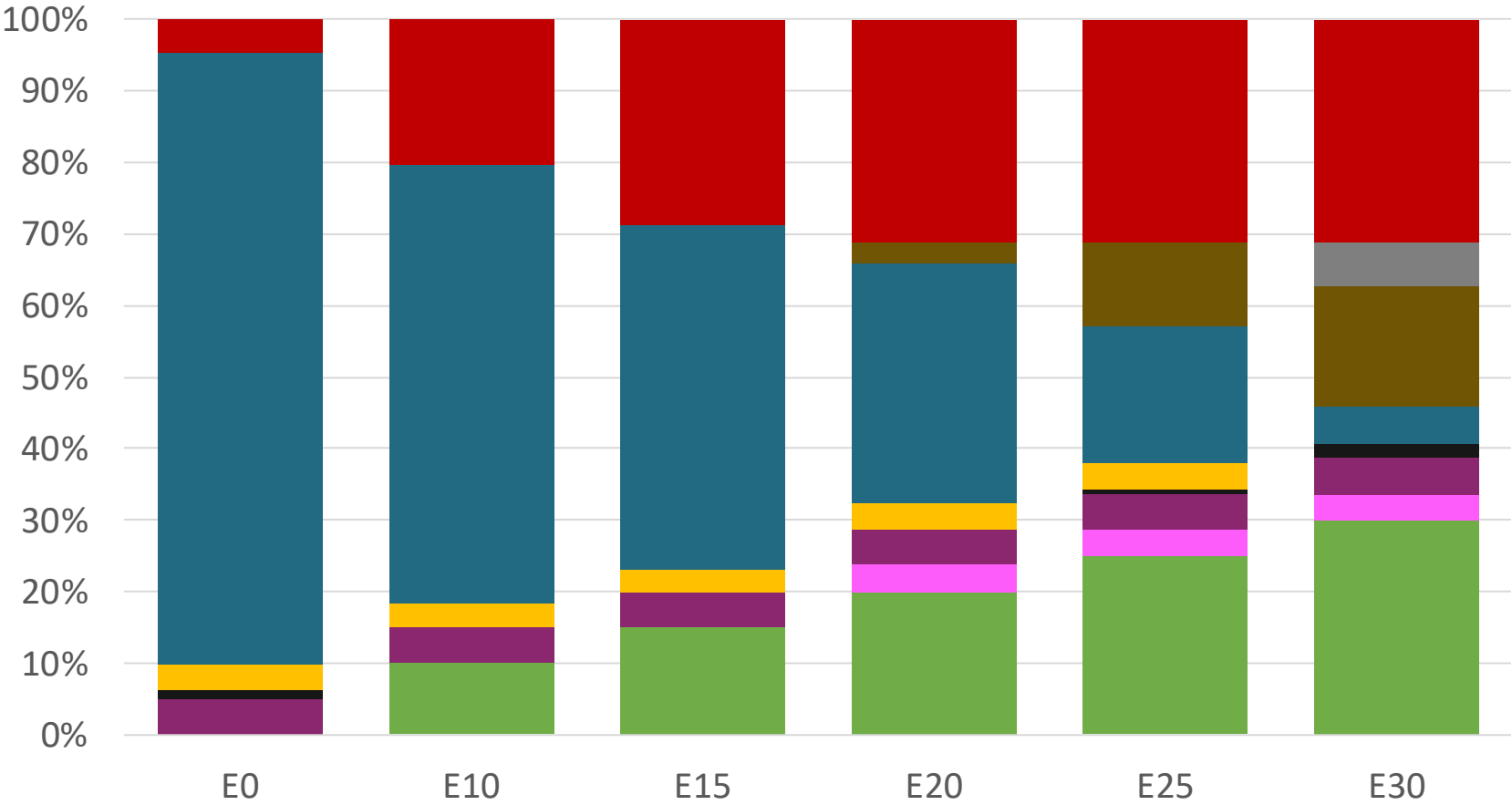
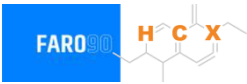


Blending Components - Costa Rica



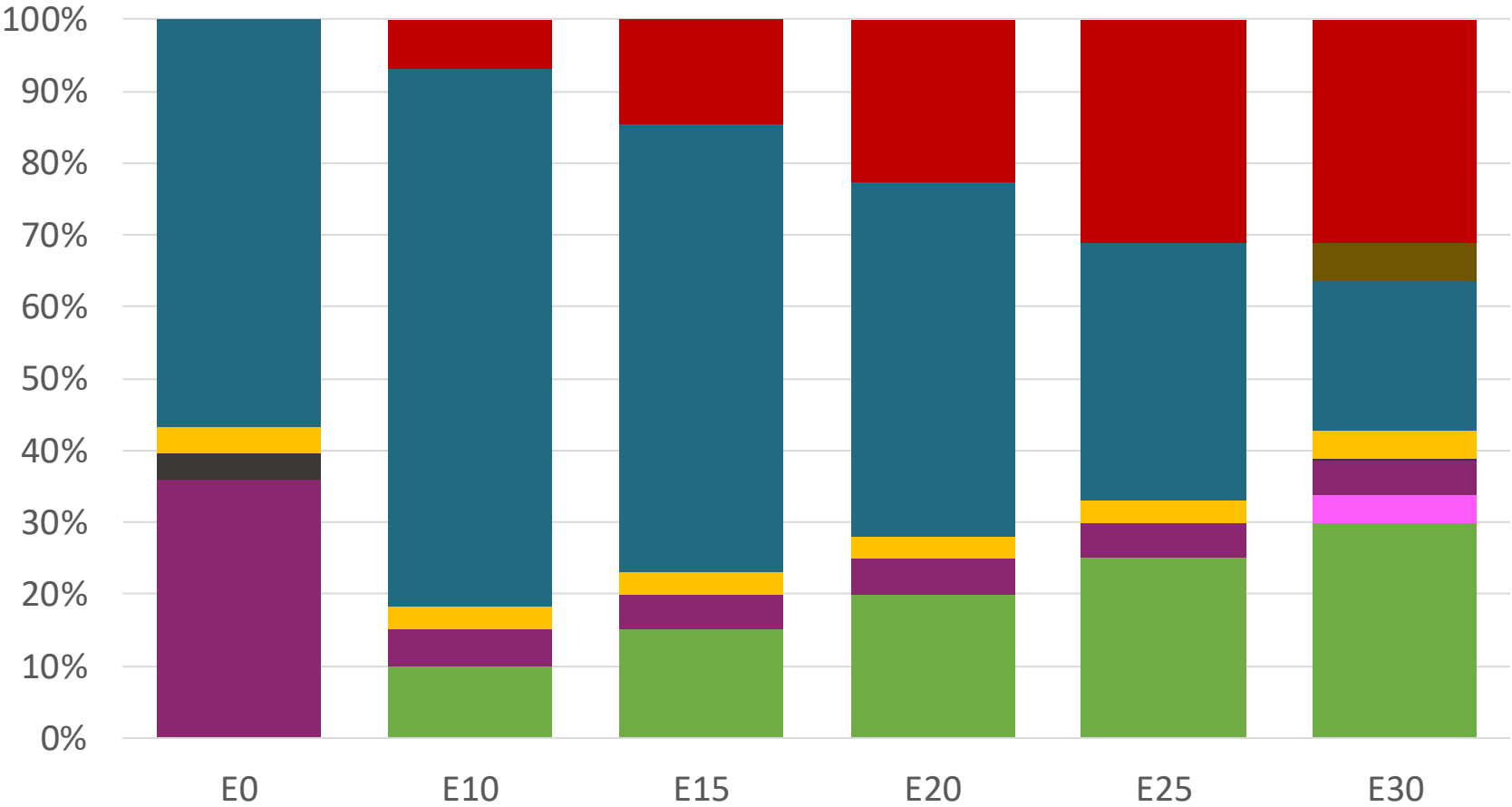
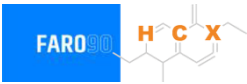
Source: Faro90

Costa Rica – Regular – Constant Octane



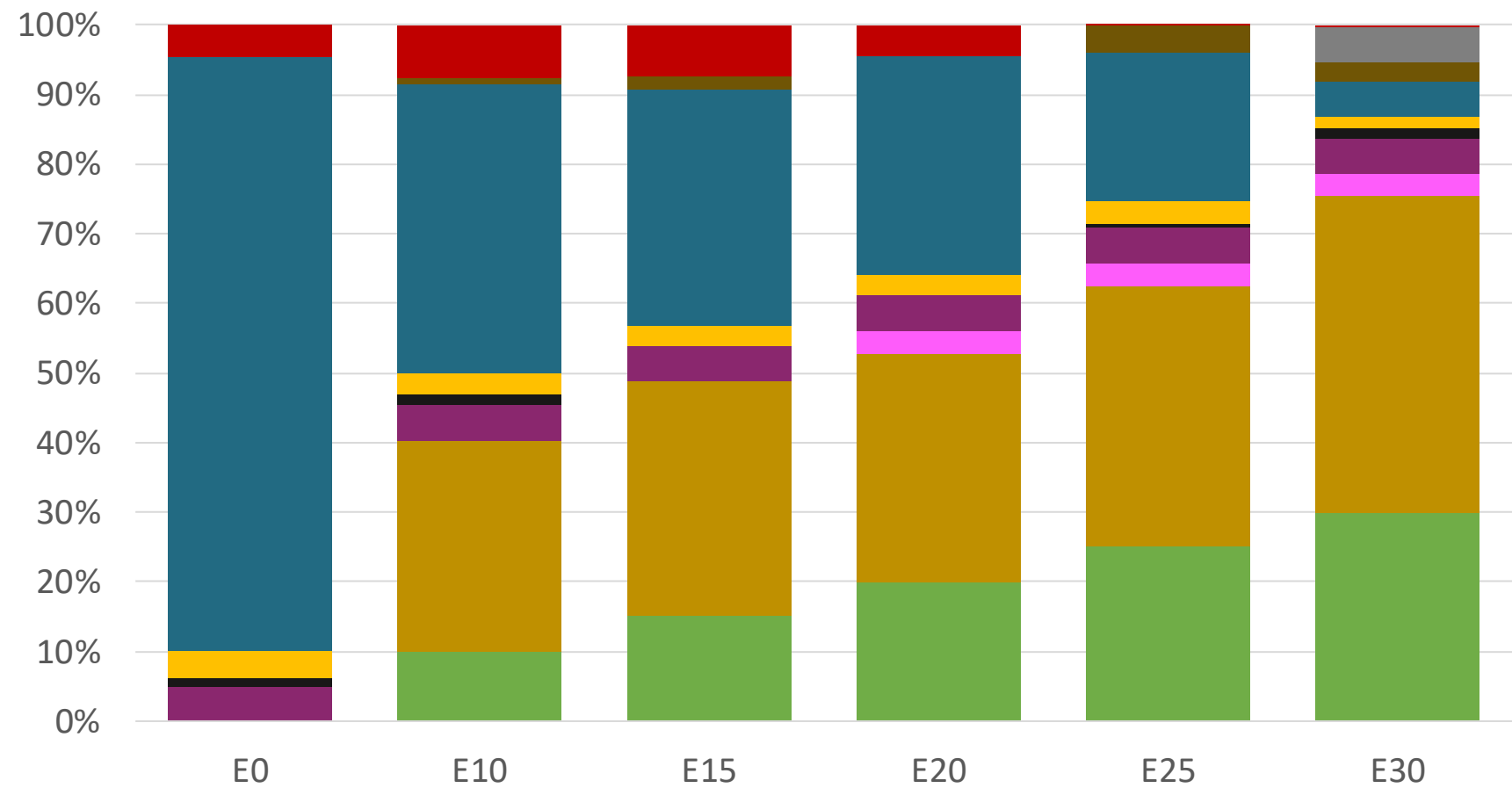
Octane (RON)	91.0	91.0	91.0	91.0	91.0	91.0
Price (USD/gal)	\$ 2.286	\$ 2.028	\$ 1.884	\$ 1.737	\$ 1.586	\$ 1.468

Costa Rica – Premium – Constant Octane



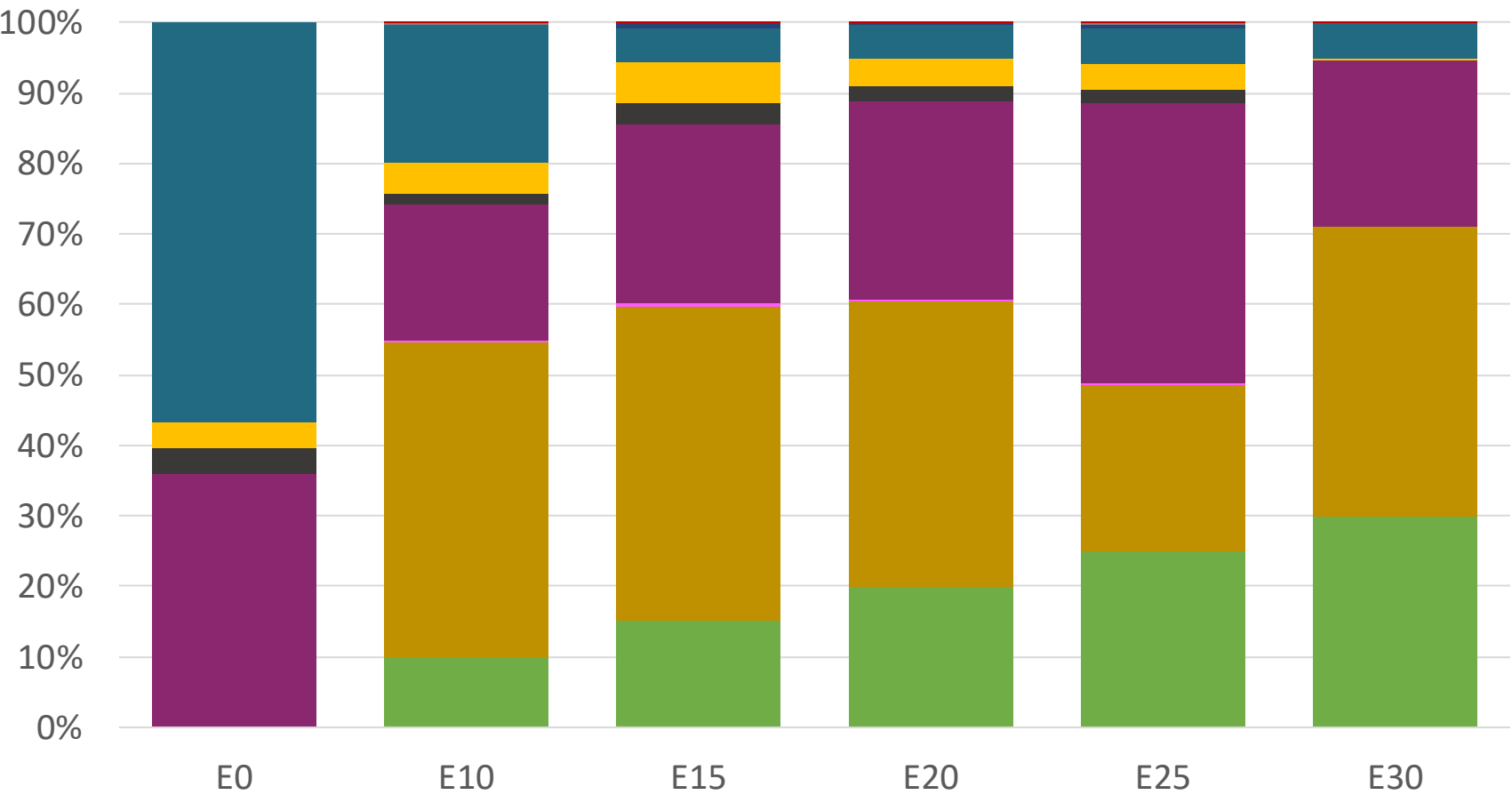
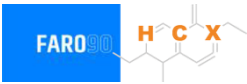
Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (USD/gal)	\$ 2.536	\$ 2.233	\$ 2.099	\$ 1.957	\$ 1.807	\$ 1.659

Costa Rica – Regular – Increased Octane



Octane (RON)	91.0	94.6	96.6	99.1	101.2	102.4
Price (USD/gal)	\$ 2.286	\$ 2.286	\$ 2.286	\$ 2.286	\$ 2.286	\$ 2.286

Costa Rica – Premium – Increased Octane



Octane (RON)	95.0	98.0	100.3	102.6	104.9	105.8
Price (USD/gal)	\$ 2.536	\$ 2.536	\$ 2.536	\$ 2.536	\$ 2.536	\$ 2.536

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

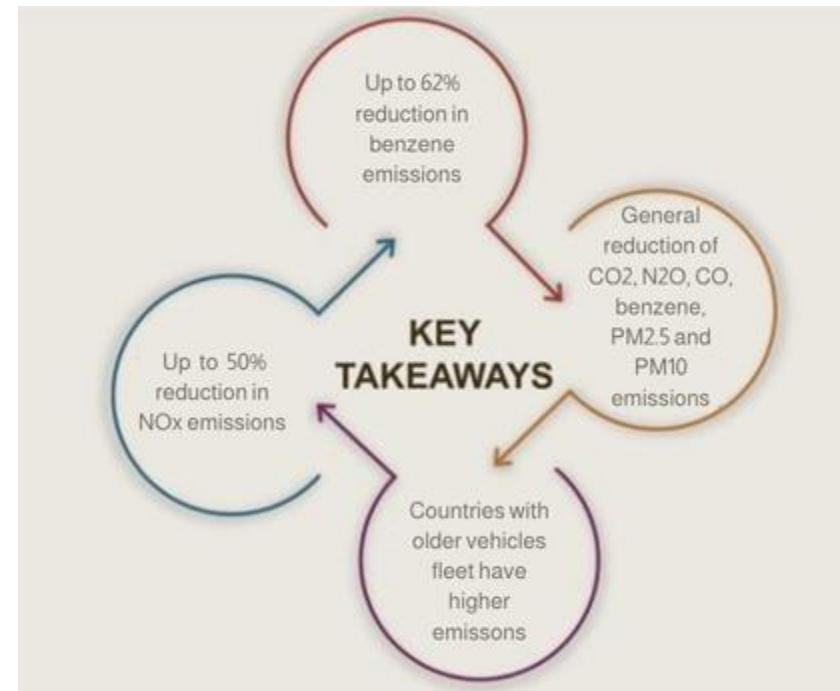
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

*Source: Bureau of transportation statistics.

Main Results



Gasoline Vehicular Fleet in Costa Rica

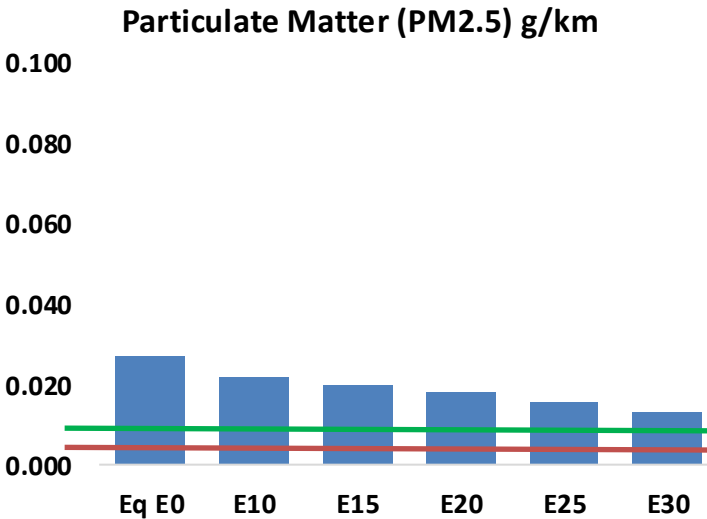
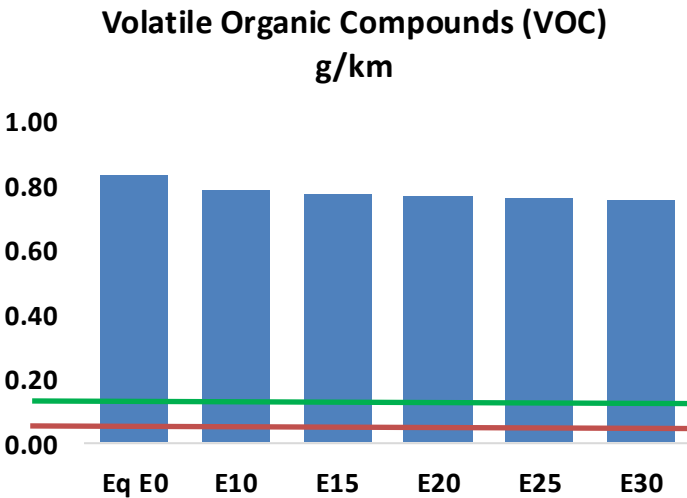
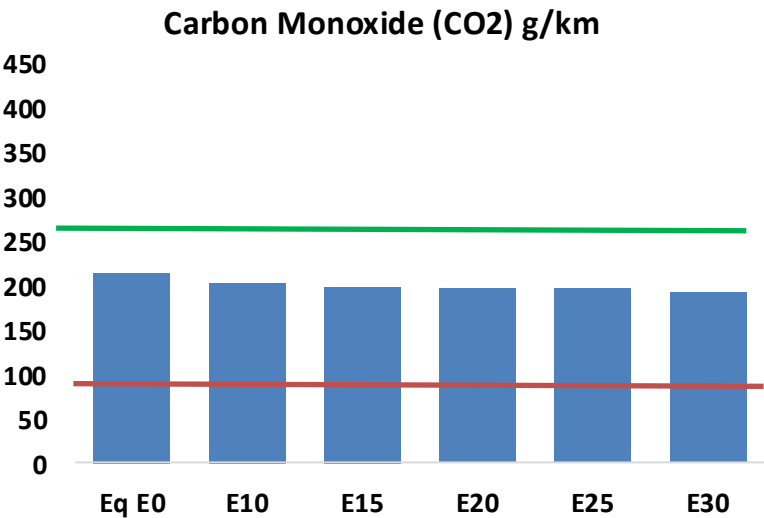
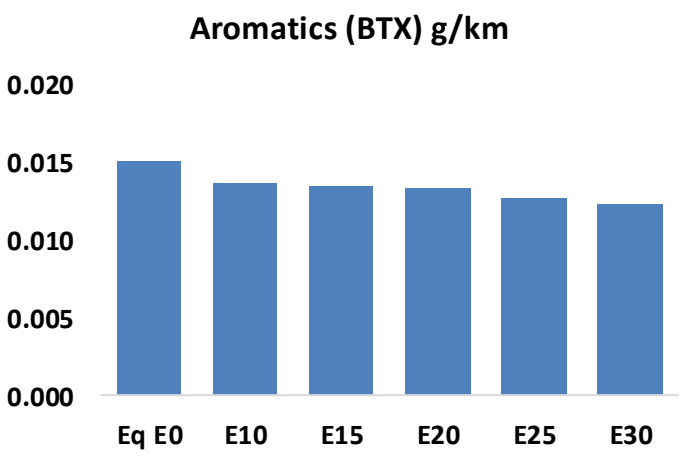
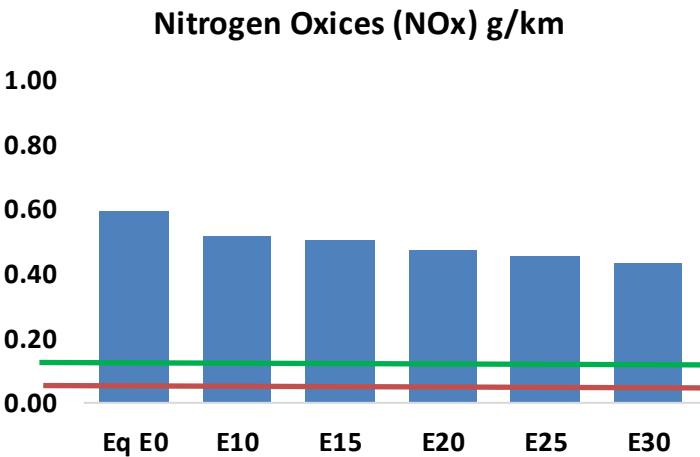
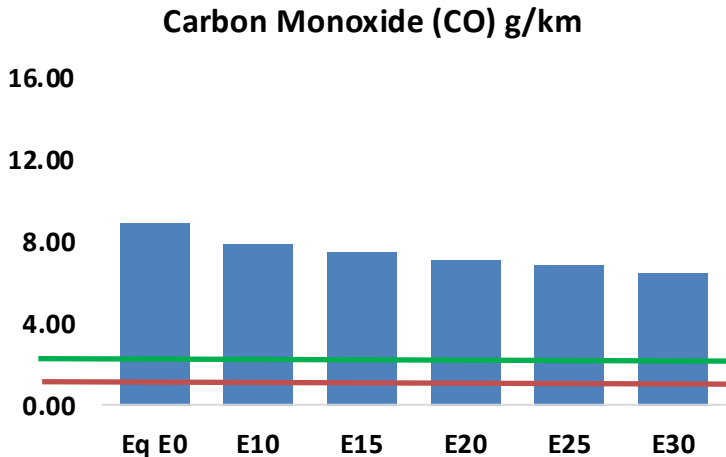
Type	Motor	Exhaust	Recuperator	Age	Number of Vehicles					
Auto/Light-Medium	Multi FI-Pt	Euro V	PCV Tank	>4years, >80 mkm						
				4-8 years, 80-160 mkm	177,395					
				<8 years, < 160 mkm	201,973					
		Euro IV		<8 years, < 160 mkm	135,031					
		Euro III		<8 years, < 160 mkm	132,105					
		3-Ways		<8 years, < 160 mkm	125,379					
					278,379					
Small Engines	FI 4-cycles	Euro III	PCV	>4years, >80 mkm	52,143					
				4-8 years, 80-160 mkm	29,684					
		Euro II		4-8 years, 80-160 mkm	29,684					
				<8 years, < 160 mkm	197,199					
Trucks	FI	Euro V	PCV	>4years, >80 mkm	36,587					
				4-8 years, 80-160 mkm	41,656					
				<8 years, < 160 mkm	27,850					
		Euro IV		<8 years, < 160 mkm	27,246					
		Euro III		<8 years, < 160 mkm	25,859					
		3-Ways		<8 years, < 160 mkm	57,415					

Vehicular Fleet: **1,575,585**
Average Age: **11 years**
Motorcycles: **19.6%**

Source: INEC , Faro 90

Costa Rica – Vehicular Emissions

United States
Euro 6



Costa Rica – Vehicular Emissions

Vehicular Emissions (kg/day)	E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	497,036	468,122	439,208	417,328	396,144	380,107	359,765	-12%	-20%
CO2	11,924,378	11,617,751	11,328,159	11,100,403	10,987,701	10,923,615	10,722,262	-5%	-8%
VOC	46,603	45,283	43,962	43,321	42,880	42,602	42,014	-6%	-8%
NOx	32,987	30,458	28,763	27,921	26,541	25,470	24,202	-13%	-20%
SOx	137	126	116	107	99	90	80	-15%	-28%
NH3	4,183	4,142	4,100	4,120	4,166	4,199	4,226	-2%	0%
N2O	166	165	164	165	169	170	172	-1%	2%
THC	9,847	9,847	9,847	9,995	10,211	10,398	10,576	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	378	368	357	352	348	346	341	-6%	-8%
Acetaldehyde	784	899	1,319	2,009	2,738	3,128	3,696	68%	249%
Formaldehyde	2,382	2,316	2,700	3,170	3,322	3,674	4,000	13%	39%
Benzene	838	805	763	751	745	712	689	-9%	-11%
PM2.5	701	637	573	517	462	404	342	-18%	-34%
PM10	790	718	646	583	521	455	386	-18%	-34%

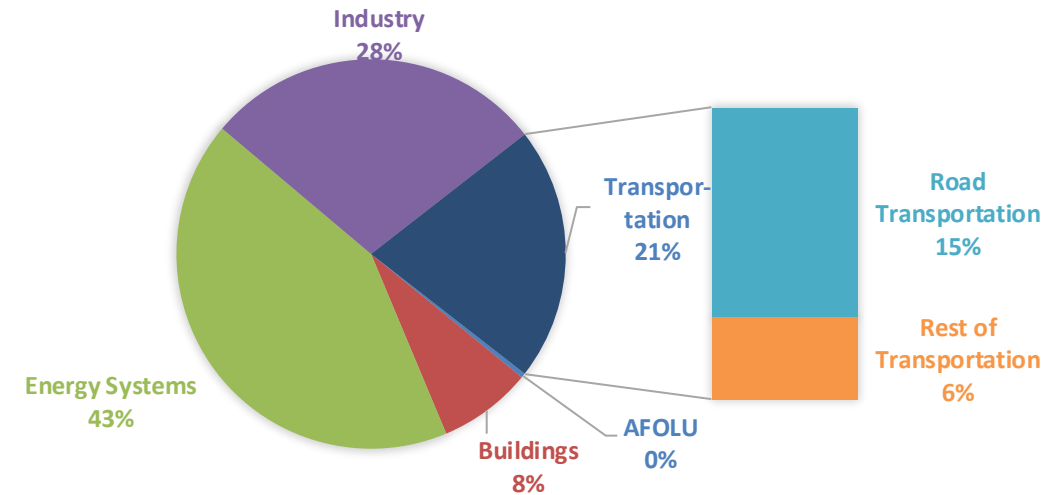
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

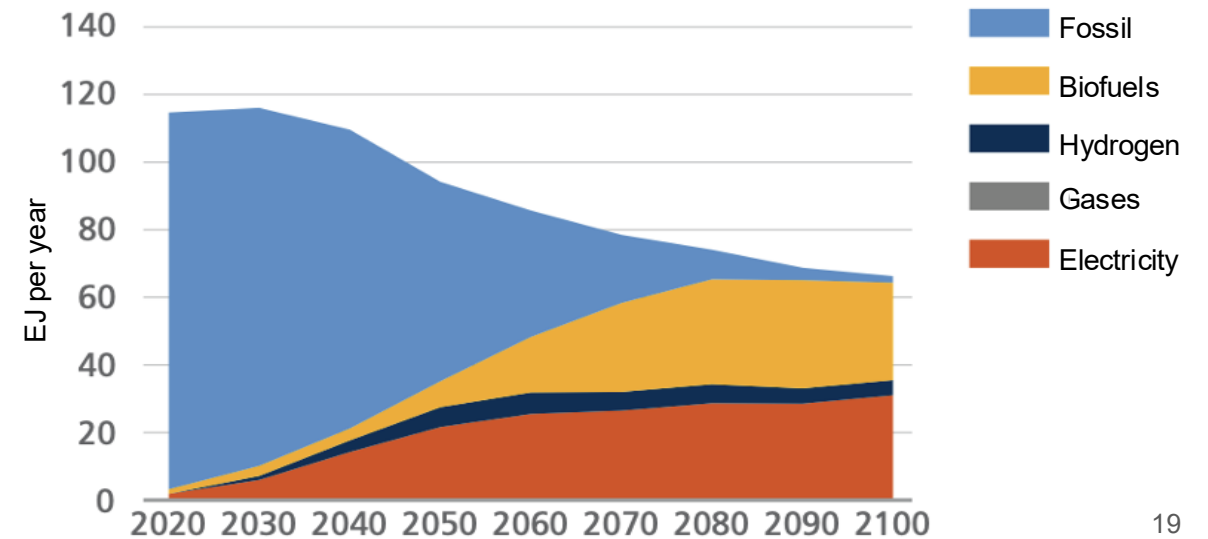
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

Current GHG Global Emissions Distribution

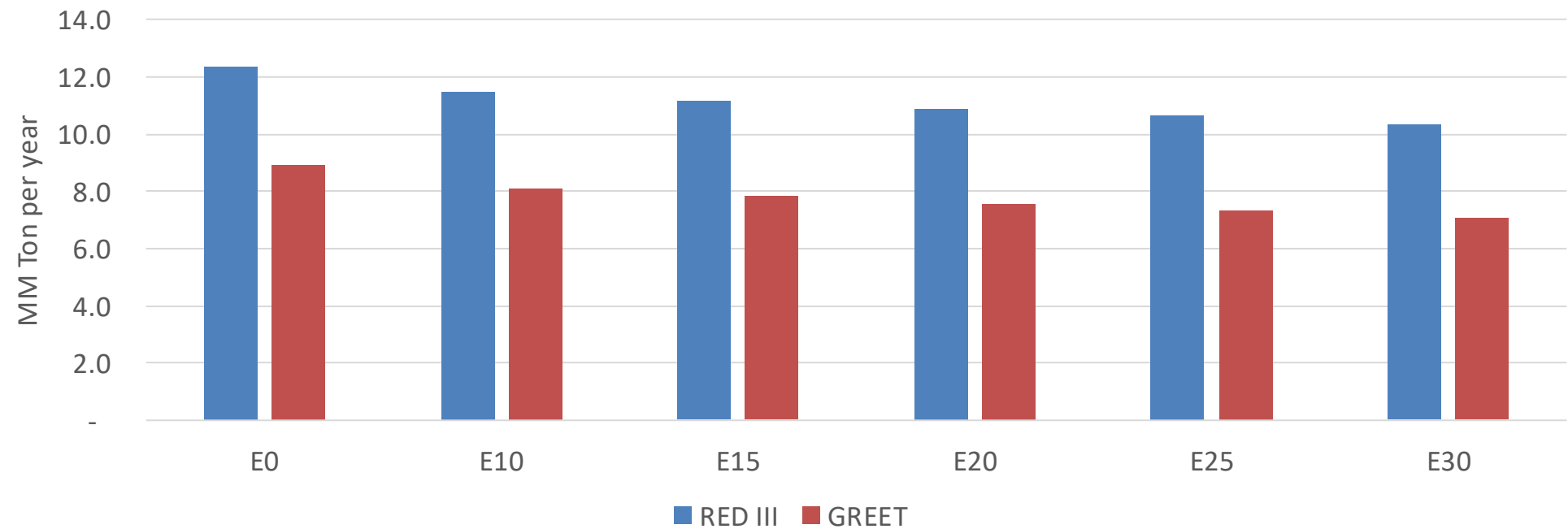


IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Costa Rica – Gasoline Vehicle Fleet GEI Emissions

Life Cycle GHG Emission - Gasoline Vehicles



Country Emissions 2024 (MMT/year)	17.1
2035 Target (MMT/year)	8.5
Delta	8.6

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	9.2%	11.9%	14.9%	17.3%	20.6%
RED II/III	0.0%	7.2%	9.4%	11.8%	13.7%	16.3%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	9.5%	12.3%	15.4%	17.9%	21.3%
RED II/III	0.0%	10.3%	13.5%	16.9%	19.7%	23.3%